

Smart Metro Ticket Booking System

Project Development Phase

2. Performance Testing

1. Purpose

This document sets out the performance testing approach for the metro digital ticketing solution on ServiceNow. Performance testing confirms the solution meets the response time, scalability, and availability targets defined in the Non-Functional Requirements, particularly under peak commuter load.

2. Objectives

- Validate that catalog booking, fare calculation, payment processing, and QR generation meet defined response time targets under expected load.
- Confirm the solution supports concurrent bookings across all stations during peak commute hours without degrading.
- Identify system bottlenecks (application, integration, or infrastructure) before go-live.
- Verify graceful degradation and recovery behaviour under stress and spike conditions.
- Confirm AFC gate validation and reporting dashboards stay responsive under sustained load.

3. Types of Performance Tests

Test Type	Purpose
Load Testing	Verify system behaviour under expected normal and peak booking volumes.
Stress Testing	Determine the breaking point by pushing load beyond expected peak capacity.
Soak Testing	Validate stability and resource usage over an extended period of sustained load.
Spike Testing	Assess system response to sudden bursts of bookings, e.g., during promotions.
Scalability Testing	Confirm the system scales as new stations, routes, or channels are added.

4. Performance Testing Process

Performance testing follows a structured six-step cycle, moving from requirement analysis through execution, monitoring, tuning, and final sign-off, with a loop back to execution after any performance tuning.

Figure 1 — Performance Testing Process Flow

5. Performance Test Scenarios

Scenario ID	Scenario Description	Test Type
PERF-SC-01	500 concurrent passengers submitting catalog booking requests during peak hour.	Load
PERF-SC-02	Simultaneous fare calculation requests across multiple stations and zones.	Load
PERF-SC-03	Concurrent payment transactions across UPI, card, and wallet gateways.	Load
PERF-SC-04	Bulk QR code generation immediately following a wave of successful payments.	Load / Spike
PERF-SC-05	Sustained booking traffic over an 8-hour operational window.	Soak
PERF-SC-06	Booking volume increased 2x beyond expected peak to identify breaking point.	Stress
PERF-SC-07	Sudden spike in bookings during a promotional fare offer.	Spike

6. Performance Targets vs Sample Results

The chart below compares target response-time thresholds (from the Non-Functional Requirements) against sample results from an early test cycle, to show how results will be tracked once full performance testing runs.

Figure 2 — Response Time: Target vs Sample Test Results

7. Performance Metrics & Targets

Metric	Target
Catalog form load time	Within 3 seconds under normal load.
Fare calculation response time	Within 2 seconds of source/destination selection.
Payment processing time	Within 5 seconds for UPI/card/wallet transactions.
QR code generation time	Within 5 seconds of payment confirmation.
AFC gate validation time	Within 1.5 seconds per scan.
Concurrent user support	Peak-hour concurrent bookings across all stations without degradation.
System availability	99.5% uptime during operational hours.
Error rate under peak load	Less than 1% failed transactions during peak load testing.

8. Test Environment & Tools

- Dedicated ServiceNow performance/UAT instance mirroring production configuration.
- Load generation tools capable of simulating concurrent catalog, payment, and QR requests.
- Monitoring tools for application, integration, and infrastructure-level metrics (CPU, memory, API latency, queue depth).
- Simulated AFC gate endpoints to test gate validation throughput without physical hardware.

9. Risks & Mitigation

Risk	Mitigation
Payment gateway throttling under load	Coordinate load test windows with the payment vendor and use sandbox rate limits appropriately.
QR generation API becomes a bottleneck	Test the QR API independently at higher concurrency to isolate its capacity limits.
Test environment not representative of production	Align environment sizing and configuration with production before executing tests.
Insufficient monitoring during test execution	Enable detailed monitoring and logging before each performance test cycle.

10. Exit Criteria

- All performance test scenarios have been executed across load, stress, soak, and spike conditions.
- All response-time metrics meet or exceed the defined targets.
- No unresolved Critical/High performance defects remain.
- System demonstrates stable behaviour and recovery under sustained and peak load.
- Performance test summary report is reviewed and signed off by IT Admins and the Metro Operations Team.